Common Laplace Transform Pairs

$f(t)$	$\Leftrightarrow$	$F(s)$	$f(t)$	$\Leftrightarrow$	$F(s)$
$\delta(t)$		1	$\sin(bt)$		$\frac{b}{s^2+b^2}$
$U(t)$		$\frac{1}{s}$	$\cos(bt)$		$\frac{s}{s^2+b^2}$
t		$\frac{1}{s^2}$	$e^{-at}\sin(bt)$		$\frac{b}{(s+a)^2+b^2}$
t^k		$\frac{k!}{s^{k+1}}$	$e^{-at}\cos(bt)$		$\frac{s+a}{(s+a)^2+b^2}$
e^{-at}		$\frac{1}{s+a}$			
te^{-at}		$\frac{1}{(s+a)^2}$			
$\frac{1}{(k-1)!}t^{k-1}e^{-at}$		$\frac{1}{(s+a)^k}$			

- EE2023: [Laplace Transform and Properties Tables]

Laplace Transform Properties

	Time Function	Laplace Transform	Comments
-	$f(t)$	$F(s)$	Transform pairs
1	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$	Superposition
2	$f(t - t_0)$	$e^{-st_0} F(s)$	Time delay $(t_0 \geq 0)$
3	$f(at)$	$\frac{1}{\lvert a \rvert} F\left(\frac{s}{a}\right)$	Time scaling
4	$e^{-at} f(t)$	$F(s + a)$	Shift in frequency
5	$f^n(t)$	$s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^k(0^-)$	Differentiation
6	$\int_0^t f(\tau) d\tau$	$\frac{1}{s} F(s)$	Integration
7	$f(t) * g(t)$	$F(s)G(s)$	Convolution
8	$tf(t)$	$-\frac{d}{ds} F(s)$	Multiplication by time
9	$f(0^+)$	$\lim_{s \to \infty} sF(s)$	Initial value theorem
10	$\lim_{t \to \infty} f(t)$	$\lim_{s \to 0} sF(s)$	Final value theorem

Laplace Transform and Properties Tables

Time-domain functions, $x(t)$		s-domain functions, $X(s)$
Unit Impulse	$\delta(t)$	**1**
Unit Step	$u(t)$	$1/s$
Ramp	$t\,u(t)$	$1/s^2$
n^th^ order Ramp	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
Damped Ramp	$t\exp(-\alpha t)u(t)$	$1/(s+\alpha)^2$
Exponential	$\exp(-\alpha t)u(t)$	$1/(s+\alpha)$
Cosine	$\cos(\omega_o t)u(t)$	$s/(s^2+\omega_o^2)$
Sine	$\sin(\omega_o t)u(t)$	$\omega_o/(s^2+\omega_o^2)$
Damped Cosine	$\exp(-\alpha t)\cos(\omega_o t)u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega_o^2}$
Damped Sine	$\exp(-\alpha t)\sin(\omega_o t)u(t)$	$\frac{\omega_o}{(s+\alpha)^2+\omega_o^2}$

- unit step, u(t), is multiplied to all the signals as Laplace transform is defined to be unilateral

	Time-domain	***s*-domain**
Linearity	$\alpha x_1(t)+\beta x_2(t)$	$\alpha X_1(s)+\beta X_2(s)$
Time shifting	$x(t-t_o)u(t-t_o)$	$\exp(-st_o)X(s)$
Shifting in the s-domain	$\exp(s_o t)x(t)$	$X(s-s_o)$
Time scaling	$x(\alpha t)$	$\frac{1}{\lvert\alpha\rvert}X\left(\frac{s}{\alpha}\right)$
Integration in the time-domain	$\int_{0^-}^{t} x(\zeta)d\zeta$	$\frac{1}{s}X(s)$
Differentiation in the time-domain	$\frac{dx(t)}{dt}$	$sX(s)-x(0^-)$
	$\frac{d^n x(t)}{dt^n}$	$s^n X(s)-\sum_{k=0}^{n-1} s^{n-1-k}\left.\frac{d^k x(t)}{dt^k}\right\rvert_{t=0^-}$
Differentiation in the s-domain	$-tx(t)$	$\frac{dX(s)}{ds}$
	$(-t)^n x(t)$	$\frac{d^n X(s)}{ds^n}$
Convolution in the time-domain	$\int_{-\infty}^{\infty} x_1(\zeta)x_2(t-\zeta)d\zeta$	$X_1(s)X_2(s)$
Initial value theorem	$x(0^+)=\lim_{s\to\infty} sX(s)$	
Final value theorem	$\lim_{t\to\infty} x(t)=\lim_{s\to 0} sX(s)$	

- Two additional for systems: